

Like only used with characters and datatype not with numeric type

* TO_DURATION → Casts a String representation of a timestamp into a number of milliseconds

* TO_DOUBLE → Casts a String representation of an Integer as a double

* TO_INTEGER → Casts TRUE/FALSE to 1/0;

* SUBSTR

* INSTR

* REPLACE

* TRANSLATE → This function is used to encrypt characters.

* Soundex :- This function is used to check pronunciation rather than exact characters

Select ename from emp where Soundex(ename) = Soundex('Smith');

* RPAD

Select rpad(ename, '*', 10) from emp;

* LPAD

* LTRIM, RTIM, TRIM

* Concat

Select concat(concat(ename, 'is a'), Job) from emp;

- * Character functions accept characters inputs and can return either characters or numbers values as output
- * When a character function returns a character value, that value is always of type VARCHAR2 with the following two exceptions: UPPER and LOWER.

* Single Row function

NVL (exp1, exp2)

NVL2 (exp1, exp2, exp3);

DECODE()

COALESCE() :- It examines the first expression, if the first expression is not null, it returns that expression; otherwise, it does a COALESCE of the remaining functions.

* The advantage of the COALESCE() function over the NVL() function is that the COALESCE function can take multiple alternative values.

* NULLIF() : It compares two expressions. If they are equal, the function returns null. If they are not equal, the function returns the first expression.

* LNNVL() :- Used in where clause.

* NANVL() : Used only for floating-point numbers of type BINARY_FLOAT or BINARY_DOUBLE

Implicit Data type Conversion

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* A VARCHAR2 or CHAR value can be implicitly converted to NUMBER or DATE type value.

* Similarly NUMBER or DATE type value can be automatically converted to character data by ORACLE SERVER.

NOTE:- Implicit interconversion happens only when the character represents the a valid number or date type value respectively.

EXPLICIT DATA TYPE CONVERSION

TO_CHAR → It is used to typecast a numeric or date input to character type with a format model (optional)

TO_NUMBER → It converts a character value to a numeric datatype. If the string being converted contains nonnumeric character, the function returns an error.

TO_DATE :- The function takes character values as input and returns formatted date equivalent of the same. It allows users to enter a date in any format, and then it converts the entry into the default format used by Oracle 11g.

Create Role Role1

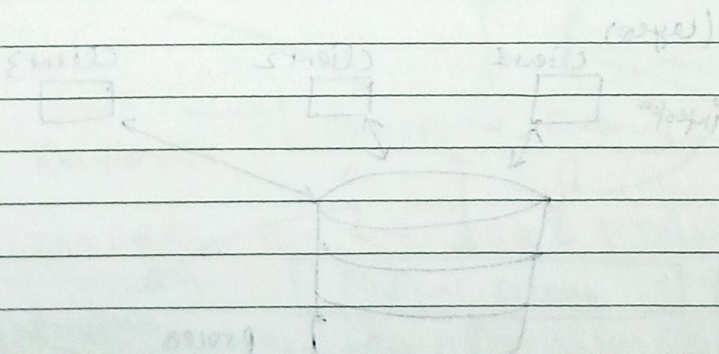
role-tab-privs, user-sys-privs, role-tab-privs

* user-tab-privs-made, user-tab-privs-recd
role-tab-privs

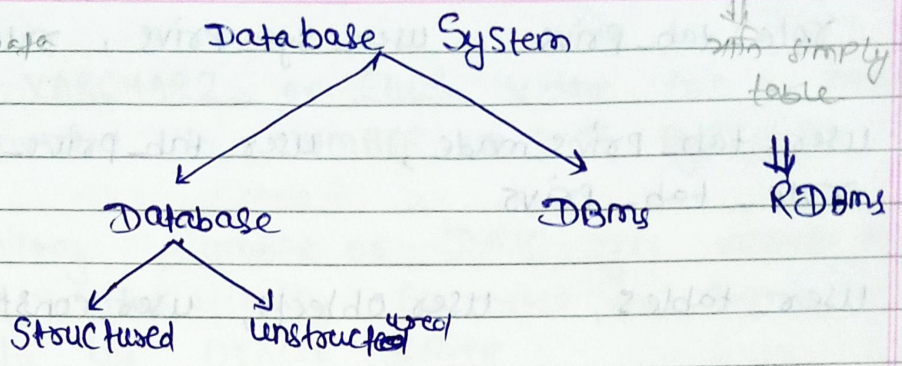
* user-tables, user-objects, user-constraints

* Select trunc(199.95, -2) from dual;

* Select Round(125.315, 3) ''



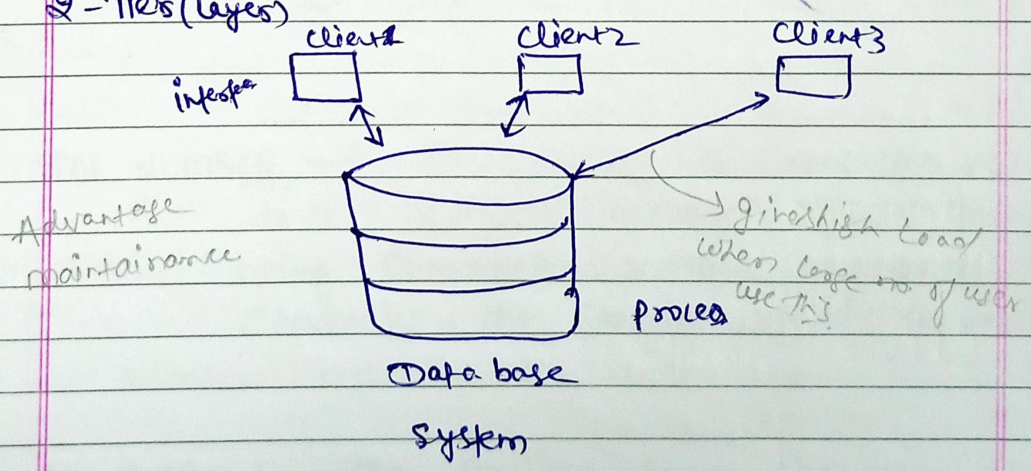
collection of
Related Data



File System vs DBMS.

- 1) 1 KB but get 25 GB. Searching is slow
- 2) ~~File~~ Attributes
3. Concurrency.
- 4) Security → DBMS provide Role based Security
- 5) Redundancy.

2-Tiers (Layers)

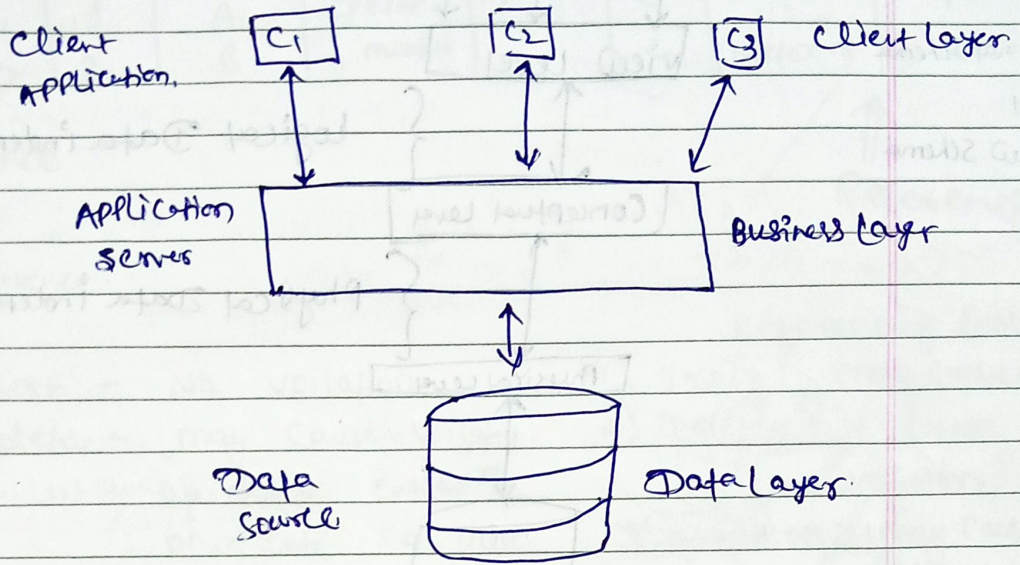


fail when large no. of user (scalability.)
→ Security

Schema $\rightarrow$ Logical Representation.

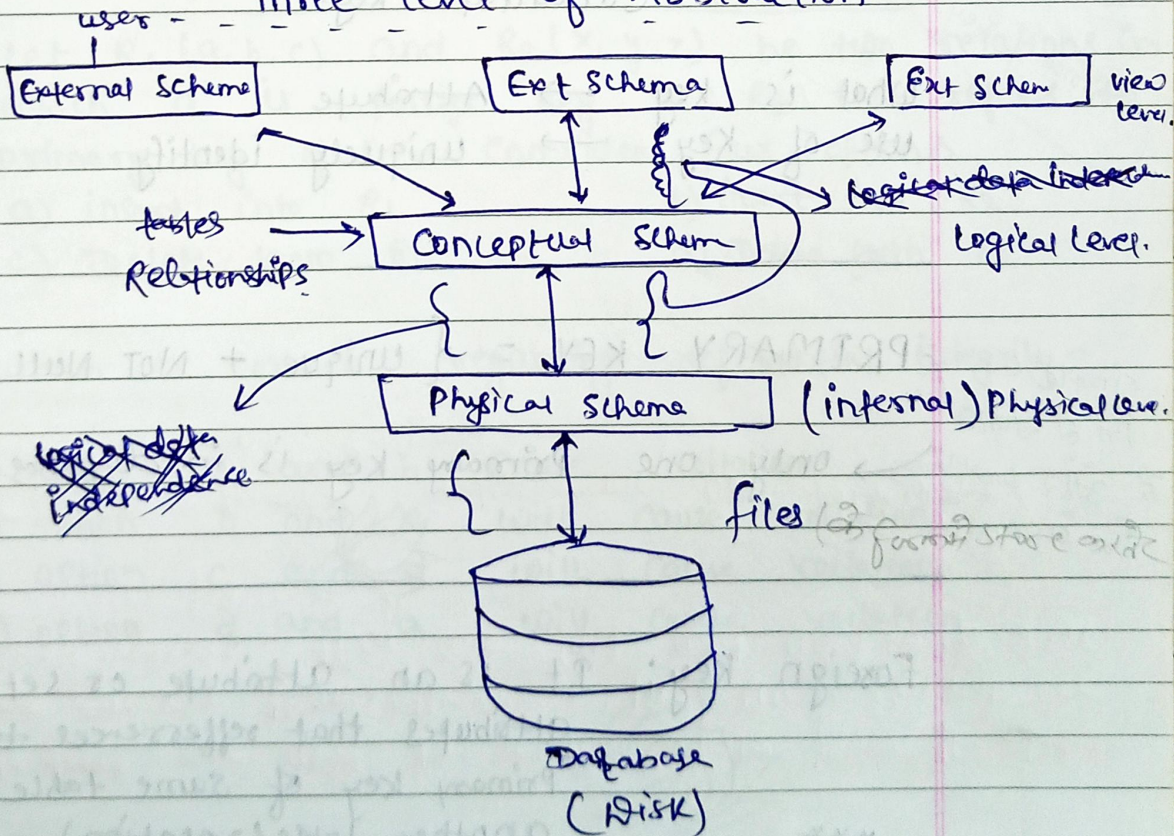
3-Tier

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Solve all 2-Tier problem.

Three level of Abstraction.



Physi

logical schema and Conceptual Schema both are same (no difference)

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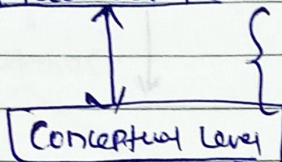
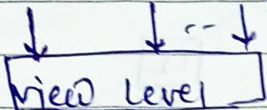
Data independence

External Schema

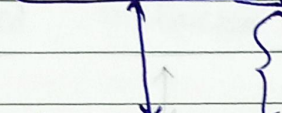
वर्तमान

view Schema

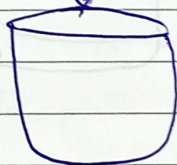
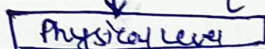
user user user



Logical Data independence



Physical data independence



↓ does not affect the
(conceptual level)
1) Storage
2) D.S. (Data Structure)
3) Index

Candidate key

what is key → Attribute

use of key → uniquely identify.

PRIMARY KEY = { unique + NOT Null }

→ only one Primary key is in database.

Foreign key: It is an attribute or set of attributes that references to Primary key of same table or another table (relation).

→ maintaining referential Integrity.

f.k Primary Key फि हो सका है

f.k Same table में फि हो सका है और दूसरी table में फि हो सकता है

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PK = 1007

fk.

RollNo.	name	address
1	A	Delhi
2	B	Mumbai

CourseId	Course name	RollNo.
C ₁	DBMS	1
C ₂	Networks	2

Reference

Referencing

Reference.

Referencing table.

- 1) Insert - No violation.
- 2) Delete - may Cause violation.
on delete cascade.
on delete set Null.
on delete No Action
- 3) update → may cause violation.

- 1) Insert: may Cause violation.
- 2) Delete: not Cause any violation.
- 3) update: may Cause violation.

Ques) Let $R_1(a, b, c)$ and $R_2(x, y, z)$ be two relations in which 'a' is foreign key in R_1 that refers to primary key of R_2 . Consider four options

- a) insert into R_1
- b) insert into R_2
- c) Delete from R_1
- d) Delete from R_2

which is correct regarding referential integrity?

- 1) option a and b Cause violation.
- 2) option b and c will Cause violation.
- 3) option c and d will Cause violation.
- ✓ 4) option d and a will Cause violation

Auto-increment allows a unique number to be generated automatically when a new record is inserted into a table.

by default
Starting value
1.

Super Key : A Super Key is a combination of all possible attributes which can uniquely identify two tuples in a table.

Auto-increment
it by one
always

Super Set of any candidate key is Super key.

Ques) $R(A_1, A_2, A_3, A_4, \dots, A_n)$ then how many Super Keys are possible?

Custom

Auto-increment = 1001

if A_1 is Candidate key.

Ans 2^{n-1}

if A_1, A_2 Both are Candidate key.

Ans $2^{n-1} + 2^{n-1} - 2^{n-2}$



if A_1, A_2, A_3, A_4 are Candidate key.

Ans. $2^{n-2} + 2^{n-2} - 2^{n-4}$

CREATE INDEX
index-name ON table-name(
column1, column2, ---)

Indexes are used to retrieve data from the database more quickly than otherwise. The users cannot see the indexes, they are just used to speed up queries.

any object which has physical existence.

Entity - Relationship Model (ER model)

it used for logical representation

Entity 

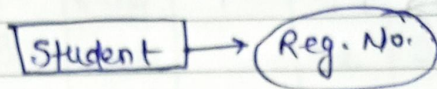
Attributes 

Relationship 

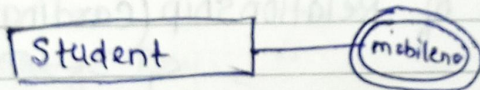
Entity or characteristic

Types of Attributes in Entity - Relationship Model.

- 1) Single Attributes
⇒ that Attribute whose value is only one.



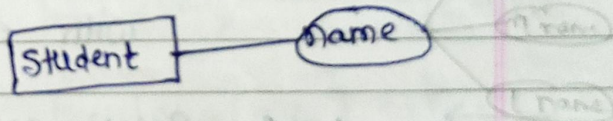
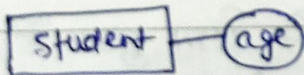
- 2) Multivalued Attributes: ⇒ more than two values



- 3) Simple vs Composite Attributes

Simple
↓
that can be broken/divide

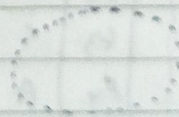
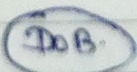
Composite
↓
composed of more than one value



- 4) Stored or Derived Attributes

↓
that attribute which can be derived (in value)

↓
Derived from any other thing



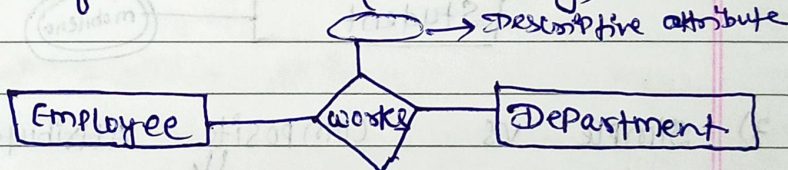
4) Key vs Non-Key Attributes
 ↓
 unique attributes may be unique or not unique.
 representation — (underline)

5) Required vs Optional Attributes (E-R)

6) Complex Attribute
 ↓
 (composite + multivalued)

Degree of Relationship (cardinality)

- 1 - 1
- 1 - M
- M - 1
- M - M (M - N)



Relationship: How the entities are connected with each other.

⇒ 1 - 1 Relationship हमें दिया हुआ है।

Eid	Ename	age	Did
E ₁	A	20	D ₁
E ₂	B	25	D ₃
E ₃	C	28	D ₂
E ₄	A	24	null
E ₅	B	25	null

Eid	Ename	age
E ₁	A	20
E ₂	B	25
E ₃	C	28
E ₄	A	24
E ₅	B	25

Eid	Did	descriptive attribute	Did	Ename	Loc
E ₁	D ₁	(Descriptive attribute)	D ₁	IT	Bang
E ₃	D ₂		D ₂	PROD	new
E ₂	D ₃		D ₃	HR	new

↓
 P.K Either Eid or Did

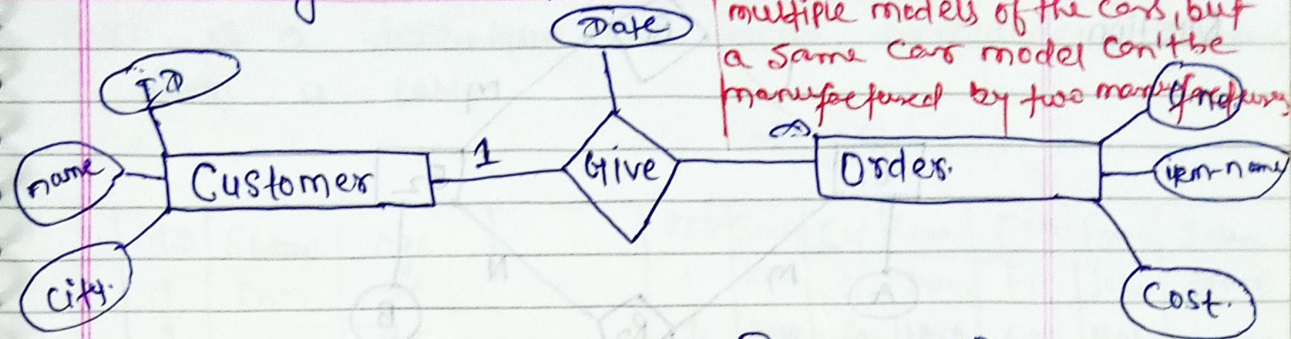
अगर हमारा P.K Eid है तो हम table को merge कर सकते हैं।

ER model is used for Designing Purpose

blue print of Database

One - Many Relationship.

A Car manufacturer makes multiple models of the cars, but a same car model can't be manufactured by two manufacturers.



1 - M Relationship हमें दिया हुआ है।

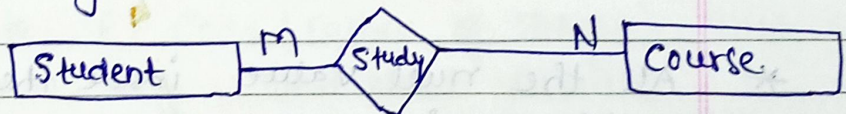
F.k			F.k		
ID	Name	City	ID	O.no.	Date
C ₁	A	Rang	C ₁	O ₁	
C ₂	B	Delhi	C ₂	O ₂	
C ₃	C	Mum	C ₂	O ₃	
C ₄	A	Mum	C ₂	O ₄	

O.no	Item no.	Cost
O ₁	Book	500
O ₂	Shoes	2000
O ₃	Shirt	1500
O ₄	Jean	2000

D.1 Primary Key.

D.2 Reduce no. of Table → merge it.

Many to many



Roll no.	Name	age
1	A	16
2	B	17
3	A	16
4	C	17

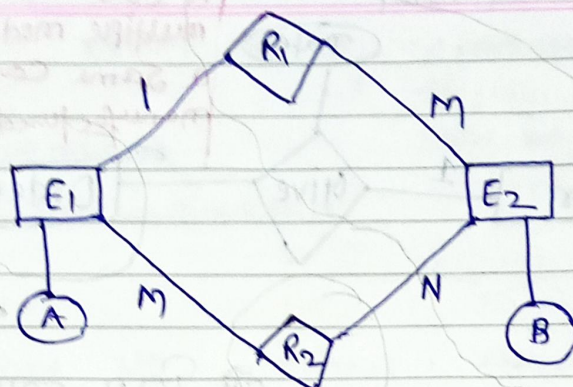
Roll no.	C-id
1	C ₁
2	C ₂
1	C ₂
2	C ₁

C-id	Name	Credit
C ₁	Maths	4
C ₂	Phy	4
C ₃	Chem	4
C ₄	Hindi	4

P.K = Composite Key Roll no. Cid

Reduce the no. of Table = No

Question.



what is the minimum no. of tables required to represent this E-R model into Relational Model?

- a) 2 b) 3 c) 4 d) 5.

Solution.

Ans B.

$$T_1 = E_1$$

$$T_2 = E_2 R_1$$

$$T_3 = R_2$$

* All the null value inside the table is treated as different internally.

Normalization

→ It is a technique to Remove or Reduce Redundancy from a table.

SID	Sname	Age
1	RAM	20
2	varun	25
1	RAM	20

Row level duplicacy.



Remove
with help of P.K.

SID	Sname	Cid	Cname	FID	Fname	Salary
1	RAM	C ₁	DBMS	F ₁	John	30000
2	RAV	C ₂	JAVA	F ₂	Bob	40000
3	Nitin	C ₁	DBMS	F ₁	John	30000
4	Amit	C ₁	DBMS	F ₁	John	30000

Column level duplicacy.

- ① Insertion Anomaly.
- ② Deletion Anomaly.
- ③ Updation Anomaly.

First Normal Form

⇒ column level dependency
एक ही column में दो से अधिक values नहीं होनी चाहिए

→ According to EF Codd (Father of DBMS) Table should not contain any multivalued Attribute.

Student

Roll no	Name	Course
1	Sai	C C++
2	Harsh	Java
3	Omkar	C DBMS

← This table is not in First Normal form.

Solution.

①

RollNo	Name	Course
1	Sai	C
1	Sai	C++
2	Harsh	Java
3	Omkar	C

Primary key: RollNo.Course

②

Rollno.	Name	course1	course2
1	Sai	C	CH
2	Harsh	Java	Null
3	Omkar	C	DBMS

Primary key

Roll no.

③

Rollno.	Name
1	Sai
2	Harsh
3	Omkar

Base table

P.K = Rollno.

Rollno	Course
1	C
1	CH
2	Java
3	C
3	DBMS

f.k

P.K = Rollno. Course

f.k = Rollno.

Finding Closure of functional dependency in DBMS

⇒ Closure method is used to find out all the candidate key in a table.

Q.1

R(ABCD)

Attributes

functional dependency

FD $\{A \rightarrow B, B \rightarrow C, C \rightarrow D\}$

$A^+ = BCDA$

$B^+ = BCD$

$C^+ = CD, D \rightarrow D$

C.K = $\{A\}$

Prime attribute = $\{A\}$

Non Prime attribute = $\{B, C, D\}$.

Prime attribute :- those attribute which are used in making of Candidate key.

Q.2. $R(ABCD)$

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$$FD = \{ A \rightarrow B, B \rightarrow C, C \rightarrow D, D \rightarrow A \}$$

$$A^+ = ABCD$$

$$CK = \{A, B, C, D\}$$

$$B^+ = BCDA$$

$$C^+ = CDAB$$

prime Attribute $\{A, B, C, D\}$

$$D^+ = DABC$$

Non prime Attribute $\{\emptyset\}$

Q.3. $R(ABCDE)$

$$FD = \{ A \rightarrow B, BC \rightarrow D, E \rightarrow C, D \rightarrow A \}$$

$$E = BDCAE$$

$$E^+ = ECA$$

$$AE = ABCED$$

$$CK = \{ \textcircled{A} E \}$$

$$\downarrow$$

$$\textcircled{D} E^+ = DEABC$$

$$CK = \{AE, \textcircled{D} E\}$$

$$\downarrow$$

$$BE^+ = BECDA$$

$$CE^+ = CE \times$$

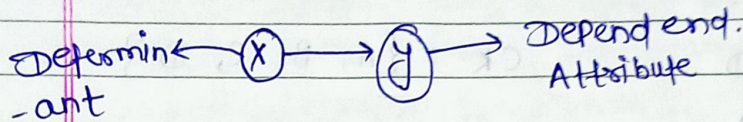
$$CK = \{AE, \textcircled{D} E, BE\}$$

Ans.

prime Attributes = $\{A, D, B, E\}$

Non-prime Attributes = $\{C\}$

Functional Dependency



X Determines Y
or

Y is Determined by X
Sid → Sname

valid ✓

Sid	Sname
1	Ranjit
2	Varun

valid ✓

Sid	Sname
1	Ranjit
1	Ranjit

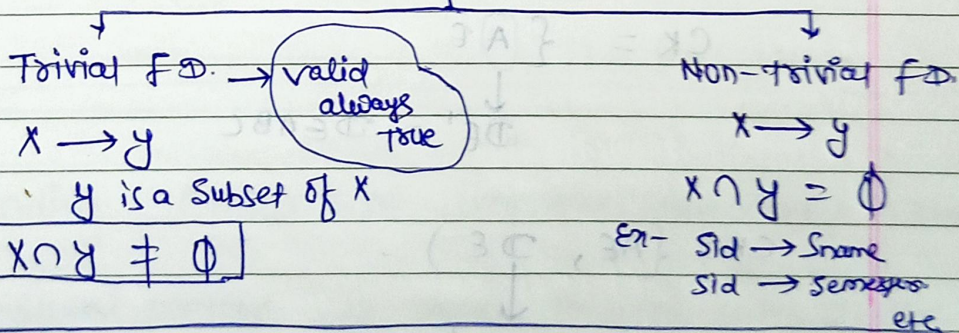
X Valid.

Sid	Sname
1	Ranjit
1	Varun

valid ✓

Sid	Sname
1	Ranjit
2	Ranjit

Functional Dependency



Properties of F.D

1. Reflexivity: if Y is a Subset of X then $X \rightarrow Y$.
2. Augmentation: if $X \rightarrow Y$ then $XZ \rightarrow YZ$.
3. Transitive: if $X \rightarrow Y$ and $Y \rightarrow Z$ then $X \rightarrow Z$.
4. UNION: If $X \rightarrow Y$ and $X \rightarrow Z$ then $X \rightarrow YZ$.
5. Decomposition: If $X \rightarrow YZ$ then $X \rightarrow Y$ and $X \rightarrow Z$.
6. Pseudotransitivity: if $X \rightarrow Y$ and $WY \rightarrow Z$ then $WX \rightarrow Z$.
7. Composition: If $X \rightarrow Y$ and $Z \rightarrow W$ then $XZ \rightarrow YW$.

Second Normal form

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According to EF Codd 2nd Normal form

- (i) Table or relation must be in 1st Normal form
- (ii) All the Non-prime attributes should be fully functional dependent on Candidate key.

Or.

There should be no partial dependency in the relation.

Customers

C-ID	Store ID	Location
1	1	Delhi
1	3	Mumbai
2	1	Delhi
3	2	Banglore
4	3	Mumbai

Partial CK

Non Prime

AB CK

either A or B

Candidate key: C-ID Store ID
Prime Attributes:

Customer ID, Store ID

Non-prime Attributes:
Location.



2nd Normal form में नहीं है।

To make them 2nd Normal form Divide the table

Customer ID	Store ID
1	1
1	3
2	1
3	2
4	3

Store ID	Location
1	Delhi
2	Banglore
3	Mumbai

2nd Normal form

candidate key

2nd Normal form

for partial dependency

L.H.S should be proper subset of C.K and R.H.S should be a Non Prime Attribute

Eg. $R(ABCDEF)$ $FD = \{C \rightarrow F, E \rightarrow A, EC \rightarrow D, A \rightarrow B\}$
 ↳ 2nd Normal form

for checking transitive dependency.

for each FD = LHS must be a CK or SK OR RHS is a prime Attribute.

नियमों पर संक्षेप

Third Normal Form

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→ table or relation must be in 2nd Normal form.

and

→ There is no transitive dependency in table.



(कोई Non Prime Attribute को Non Prime Attribute से नहीं determine कर सकता है।)

C.K = {Roll no.}

P.A = {Roll no.}

NPA = {State, City}

Rollno.	State	City
1	Punjab	Mohali
2	Haryana	Ambala
3	Punjab	Mohali
4	Haryana	Ambala

F.D → Rollno. → State ✓ valid.

State → City X not valid

Rollno. → State and State → City

Roll no. determine City through State.

अगर यहाँ पर C.A.D से P.A होता तो कोई विकल नहीं आता

↓ this is called transitive dependency.

eg. R(ABCD)

FD: AB → C, C → D

CK: AB

P.A = A, B NPA = C, D.

R = (ABCD)

FD: AB → CD, D → A

CK = {AB, DB}

Database में हमें ऐसा ही फंक्शन दिया हुआ रहता है।

BCNF (Boyce Codd Normal form)

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→ It is also said that special case of 3rd Normal form
(यानी पहले 3rd Normal form में होना चाहिए BCNF को
check करने के लिए),

eg.

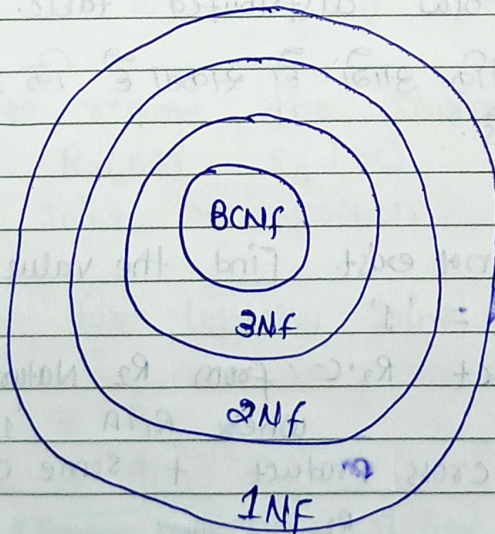
Student

Rollno.	Name	voterid	age
1	Ravi	K0123	20
2	Vasun	M012	21
3	Ravi	K786	23
4	Rahul	D286	21

C.K: { Rollno., voterid }

FD: { Rollno. → name
Rollno. → voterid
voterid → age
voterid → Rollno }

For BCNF: LHS of each FD should be
Candidate key or Super key or PK



Lossless / Lossy Join Decomposition

⇒ Whenever we normalize the table. it means we decompose the table. According to EF Codd there are two rules which have to follow

- (1) Lossless vs Lossy
- (2) Dependency Preserving Decomposition

eg. Lossless / Lossy

A	B	C
1	2	1
2	2	2
3	3	2

→ R₁(AB)

→ R₂(BC)

A	B
1	2
2	2
3	3

B	C
2	1
2	2
3	2

⇒ there is a rule of decomposition which is when decompose the table then some attribute must be common b/w distributed table.

↓
 क्योंकि आगे हो सकता है कि join करने का जरूरत पड़े.
 ↓

Suppose R is not exist. Find the value of C if the value of A = '1'

soln: Select R₂.C from R₂ Natural Join R₁ where R₁.A = '1'.

Natural Join = Cross Product + some condition.

natural join
 उसी के लिए डिस्टिग
 B की value same होगा /

R ₁		R ₂		Natural Join
A	B	A	C	
1	2	2	1	✓
1	2	2	2	✓
1	2	3	2	✗
1	2	3	2	✗
2	2	2	1	✓
2	2	2	2	✓
2	2	3	2	✗

A	B	B	C
3	3	2	1
3	3	2	2
3	3	3	2

Candidate key relation which is unique.

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After Joining the new table will be

A	B	C
1	2	1
1	2	2
2	2	1
2	2	2
3	3	2

Spurious rows

Spurious rows

data inconsistency

original table R

this is called lossy decomposition because we gain some more rows as compare to R.

Criteria for common attribute in distributed table:
Common attribute should be CK or SK of either R_1 or R_2 or Both.

So. to remove the spurious hits we take $R_1(AB)$ $R_2(AC)$ which is called lossless Joint Decomposition.

Condition for lossless Joint Decomposition

- $R_1 \cup R_2 \equiv R$
- $R_1 \cap R_2 \neq \emptyset$

Common attribute of here is:

- R_1 Candidate Key or R_2 CK or both.

Same Correlation

5th Normal form में नि
use करेंगे

Dependency preserving Decomposition.

Let $R(ABCD)$ with functional Dependencies

$$\{A \rightarrow B, B \rightarrow C, C \rightarrow D, D \rightarrow B\}$$

R is decomposed into $R_1(AB), R_2(BC), R_3(BD)$

Solution:

	$R_1(AB)$	$R_2(BC)$	$R_3(BD)$
सारा FD का (combination बनाकर के चेक करेंगे)	$A \rightarrow B \checkmark$ $B \rightarrow A \times$	$B \rightarrow C \checkmark$ $C \rightarrow B \checkmark$	$B \rightarrow D \checkmark$ $D \rightarrow B \checkmark$
(केवल Non transitive वाला है कि नहीं)	↓ Closure B का लेके लेके होंगे $B^+ = BCD$		

Union of all.

$$\begin{matrix} A \rightarrow B \\ B \rightarrow C \\ C \rightarrow B \\ B \rightarrow D \\ D \rightarrow B \end{matrix}$$

$$C^+ = CBD$$



उपर वाले से Compare करें अगर गरी
मिलनी पर Closure लेके देखें.

Finally पूरा match करेपर
dependency preserve है.

Candidate Key को Attribute होता है जो कि सारे Attributes को determine कर ले।

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Q.2

Let $R(ABCD)$ with functional dependencies
 $\{AB \rightarrow CD, D \rightarrow A\}$

→ closure

$$A^+ = A$$

$$B^+ = B$$

$$D^+ = DA$$

$$C^+ = C$$

$$BC^+ = BC$$

$$BD^+ = BDAC$$

$$CD^+ = CDA$$

Decompose $R_1(AD)$ & $R_2(BCD)$

$R_1(AD)$

$$A \rightarrow D \quad \times$$

$$D \rightarrow A \quad \checkmark$$

$R_2(BCD)$

$$B \rightarrow CD \quad \times$$

$$C \rightarrow BD \quad \times$$

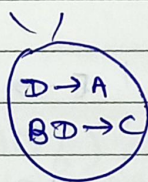
$$D \rightarrow CB \quad \times$$

$$BC \rightarrow D \quad \times$$

$$BD \rightarrow C \quad \checkmark$$

$$CD \rightarrow B \quad \times$$

union



अब $AB^+ =$ closure निकालें
 वाले को देखकर लेंगे

$$AB^+ = AB$$

यहाँ पर Dependency preserving नहीं हुआ है।

not modifiable

modifiable

Difference b/w Primary Key & Unique Key.

- Primary Key is used to identify a row ~~but~~ whereas unique key is to prevent duplicate values in a column.
- By default SQL-engine creates clustered index on primary-key if not exists and Non-Clustered index on unique-key.
- Primary key does not include Null values whereas unique-key can include null values.
- A table can have at most one P.K but can have multiple unique Key.
- Primary Key = unique key + not null constraint.

Candidate key should always be minimal $\leftarrow$ Property of $\neq$ Candidate key

Question: R(ABCDEF), Check the Highest Normal form?
FD: $\{ AB \rightarrow C, C \rightarrow DE, E \rightarrow F, F \rightarrow A \}$

Solution:

$AB^+ = ABCDEF$

$FB^+ = FBACDE$

$EB^+ = EBFACD$

$CB^+ = CBDEFA$

Step 1:

Find all CKs in Relation

CK: $\{ AB, FB, EB, CB \}$

Step 2: Write all prime Attributes $\leftarrow A$
 $\{ A, B, C, E, F \}$

Step 3: Write all Non-prime Attr.
 $\{ D \}$

	$AB \rightarrow C$	$C \rightarrow DE$	$E \rightarrow F$	$F \rightarrow A$
X BCNF	✓	X	X	X
X 3NF	✓	X	✓	✓
X 2NF	✓	X	✓	✓
INF	✓	✓	✓	✓

$\leftarrow$ by default INF होता है

Ans 1

INF चेक करते हैं कि value दे होना चाहिए

$R(ABCD EF)$ $\leftarrow$ बता नहीं सकते हैं कि single है या Composite है।

Atomic value Attribute है या नहीं बता नहीं सकते हैं।

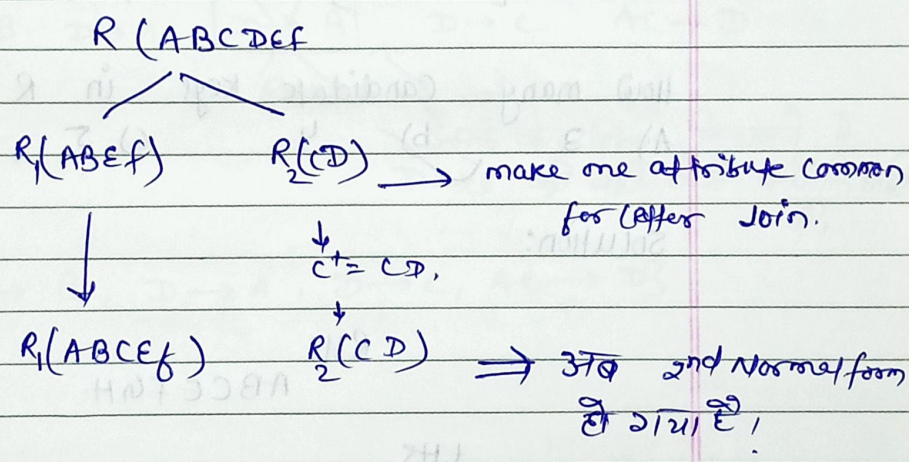
Candidate key में P.K. सि आता है।

Ques R(ABCDEF) FD: { $AB \rightarrow C$, $C \rightarrow D$, $C \rightarrow E$, $E \rightarrow F$, $F \rightarrow A$ }
 CK = { AB, FB, EB, CB }
 Prime Attribute { A, B, C, E, F }
 NPA { D }

Soln.
 for 2nd Normal form.
 $C \rightarrow D$ become PD. so it create a problem.

To make them in Second Normal form separate that Attribute which create a problem so here is (D).

कौन से Attribute
 है जो उस Attribute
 को अलग करके दिया
 जाये.

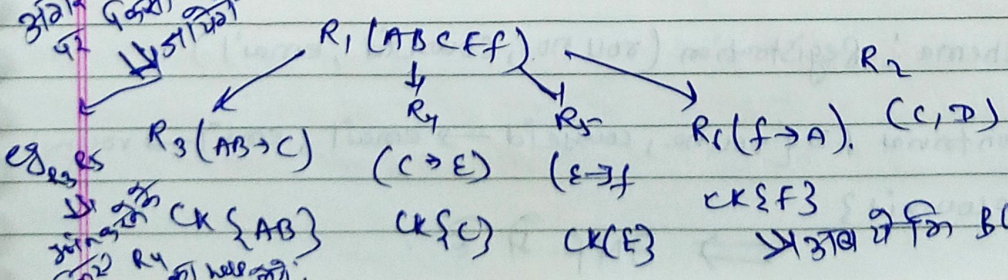


$R_1 \{ AB \rightarrow C, C \rightarrow E, E \rightarrow F, F \rightarrow A \}$ $R_2 \{ C \rightarrow D \}$
 CK { AB, FB, EB, CB } CK { C }

Join करने के लिए
 common attribute
 आता है जो कि
 join करने में
 आता है।

BCNF नहीं है।

अब ये 3rd Normal form है कि
 एम्पल हो रहा है (check करके देखो)



अब ये कि BCNF में हो गया है।

Schema $\rightarrow$ structure of a table

- Ques) Which of the following statement is true?
- A) A relation is a 3rd NF then it is always in BCNF
- B) A relation is in 2nd NF. then it is not in 1st NF.
- ✓ C) A relation is in BCNF then it is in 2NF.
- D) A relation is in 2NF then it contain PD.

Ques) Relation R has eight attributes ABCDEFGH

FD: $\{CH \rightarrow G, A \rightarrow BC, B \rightarrow CFH, E \rightarrow A, F \rightarrow EH\}$

How many Candidate keys in R

- A) 3 b) 4 c) 5 d) 6.

Solution:

RHS

ABCEFGH.

LHS

$D^+ = D$

$$\underbrace{4}_{\text{}} \left\{ \begin{array}{l} A^+ = A B C F H E G \\ E^+ = E D A B C E F G H \\ F^+ = A B C D E F G H \\ B^+ = A B C D E F G H \end{array} \right.$$

Ques.2 Schema: Registration (roll no., courseid, email)

Nontrivial FD $\{\text{roll no.}, \text{courseid} \rightarrow \text{email}, \text{email} \rightarrow \text{roll no.}\}$

CK $\{\text{roll no.}, \text{courseid}\}$

$\Rightarrow$ 3NF $\nRightarrow$ 2nd.

Q. For the following functional dependencies, find the correct minimal cover → an irreducible set of FD.

$\{A \rightarrow B, C \rightarrow B, D \rightarrow ABC, AC \rightarrow D\}$

- a) $A \rightarrow B, C \rightarrow B, D \rightarrow A, AC \rightarrow D$
- b) $A \rightarrow B, C \rightarrow B, D \rightarrow C, AC \rightarrow D$
- c) ~~$A \rightarrow BC$~~ , $D \rightarrow CA, AC \rightarrow D$
- d) $A \rightarrow B, C \rightarrow B, D \rightarrow AC, AC \rightarrow D$

Solution

Step 1

$A \rightarrow B \quad C \rightarrow B \quad D \rightarrow A \quad \boxed{D \rightarrow B} \quad D \rightarrow C \quad AC \rightarrow D$

Redundant

यहाँ इसके बाद closure निकालने में $D \rightarrow B$ को नहीं लेनी क्योंकि यह है,

Step 2

$\{A \rightarrow B, C \rightarrow B, D \rightarrow A, D \rightarrow C, AC \rightarrow D\}$

Step 3

$A \rightarrow B, C \rightarrow B, D \rightarrow A, D \rightarrow C, AC \rightarrow D$

उपसर्प नहीं कर सकते

$AC \xrightarrow{x} C \rightarrow A$
 $\downarrow$
 $\odot$

अगर माने पर C वैध है

$$\gamma = \{A \rightarrow B, B \rightarrow C, A \rightarrow C\}$$

if X covers Y $X \supseteq Y$
if Y covers X $X \subseteq Y$
then
 $X \equiv Y$

$A^+ = ABC$

↓
य से लगे।

↓
य सब x से लगे।

$B^+ = BC$

↑
x से लगे।

$A^+ = BC$
 $B^+ = BC$

So $X \equiv Y$

$x \neq y$ Ans.

$$\gamma = \{AB \rightarrow C, AB \rightarrow D, C \rightarrow D\}$$

Joins

- 1) Cross Join or Cross Product.
- 2) Natural Join.
- 3) Conditional Join
- 4) Equi Join.
- 5) self Join
- 6) Outer Join
 - Left
 - Right
 - full

Cross Product यानि एक Row को उठाओ और दूसरा table के सारे Row से multiply कर दो।

Natural Join.

→ When Common Attribute keep equal.

eg.

EMP			DEPT.		
E No.	E_name	Address	Dept No.	Name	E-No
1	Ram	Delhi	D1	HR	1
2	varun	Chd	D2	IT	2
3	Ravi	Chd	D3	MRKT	3
4	Amrit	Delhi			

Ques) find the Emp Names who is working in a department?

Select E_name from (EMP, DEPT) where EMP.Emp = DEPT.Emp

Natural Join

इसका यानि Cross Product या Join.

ये सिखा करे पर बिना Condition का max row बनायेगा.

Ans Ram, varun, Amrit

Select E_name from EMP Natural Join DEPT.